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Sub : Physics Assignment 3

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Quantum Basic

physics Assignment - 3

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Q. Explain the 4 types of transformation?

Ans. There are four types of transformations.

1. Translation

2. Scalling

3. Shearing

4. Rotation

Transformation are mathematical operations used to change the positions, size, orientation, or shape of an object. In Computer graphics and 3d geometry, transformations are commonly applied to object such as cubes, spheres and other 3d models.

1. Translation: Definition: Translation means moving an object from one position to another without changing its size or shape.

Cube example: if a cube is moved 3 units to the right, its position change but its size and shape remain the same.

2. Scalling: definition Scalling means increasing or decreasing the size of an object.

Cube ex: if the side of a cube is 2 units and we scale it by 2, its side becomes 4 units. the cube becomes bigger.

3. Rotation: Definition Rotation means turning an object around a fixed axis by a particular angle.

Cube ex: if a cube is rotated by 90° around the Z-axis, its orientation changes, but its size and shape remain the same.

4. Shearing Shearing means slanting or distorting an object by shifting its different part in a particular direction.

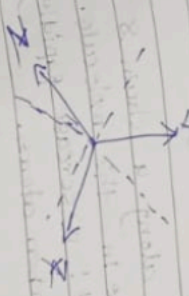
Cube ex: if shearing is applied to a cube, its slanted sides become slanted, making the cube look like a distorted or tilted box.

Q.2 Solve the following examples by applying Scaling Technique?

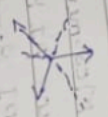
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 5x & 0 \\ 0 & 5y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} &= \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \times 3 \\ 0 \times 3 + 4 \times 2 \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \end{bmatrix} \quad x' = 9, y' = 8 \end{aligned}$$

$$\textcircled{2} P(3, 4, 2)$$



$$\textcircled{3.2} P(4, 1, 2)$$



Unit 3 : Scaling transformation.

Q.1 A point $P(4,3)$ in a 2D plane is to be scaled with scaling factors $Sx = 2$ and $Sy = 3$. Find the new coordinates P' s using matrix multiplication.

→ Given: $P(4,3)$
→ $Sx = 2, Sy = 3$

$$\Rightarrow S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \quad P = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\Rightarrow P' = S \times P = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2(4) + 0(3) & + & 0(1) \\ 0(4) + 3(3) & + & 0(1) \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \end{bmatrix}$$

$$\Rightarrow \boxed{P'(8,9)} \text{ ans.}$$

Q.2 A triangle has vertices $A(1,1), B(4,1), C(4,5)$

it is scaled about the origin with $Sx = 2$ and $Sy = 2$. Find the new vertices using matrix.

$$\Rightarrow S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$C' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$$

$$\Rightarrow \boxed{A'(2,2), B'(8,2), C'(8,10)}$$

Q.3

A 3D point $P(2,3,4)$ is scaled using $Sx = 2, Sy = 1, Sz = 3$. Using the homogeneous Sx, Sy, Sz scaling matrix, find the new point P' .

$$\Rightarrow S = \begin{bmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & Sz \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow P' = S \times P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 12 \end{bmatrix}$$

Q.4

$$\Rightarrow \boxed{P'(4,3,12)}$$

Q.5 Write the general 3D scaling matrix in homogeneous coordinates and use it to scale a cube vertex $P(1,1,1)$ uniformly by a factor of 2 in all three directions.

$$S = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= S_x = S_y = S_z = 2$$

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$P' = SXP = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$P' = (2, 2, 2) \text{ and}$$

Q.5 A 2D object is scaled about a fixed point $F(2, 2)$ instead of the origin with $S_x = 2$ and $S_y = 2$. A vertex of the object is $A(4, 4)$ using the composite matrix method find the new coordinates of A .

$$\Rightarrow \text{Given: } F = (2, 2)$$

$$A = (4, 4)$$

$$S_x = S_y = 2$$

Step 1: translate fixed point F to origin.

$$T_1 = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 3: } T_2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = T_2 S T_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -4 \\ 0 & 2 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = T_2 S T_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -4 \\ 0 & 2 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A' = (C, C) \text{ and}$$

Q.6 Find the image of point $P(3, 4)$ after reflection about the X -axis using matrix multiplication.

$$P = [3, 4]$$

$$R_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow P' = R_x P = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} [3, 4] = [3, -4]$$

Q.7 Find the image of point $P(3, 4)$ after reflection about the Y -axis using matrix multiplication.

$$P = [3, 4]$$

$$R_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$P' = R_y P = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} [3, 4] = [-3, 4]$$

$$P' = [-3, 4] \text{ and}$$

$$S = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= S_x = S_y = S_z = 2$$

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$P' = SXP = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$P' = (2, 2, 2) \text{ and}$$

Q.5 A 2D object is scaled about a fixed point $F(2, 2)$ instead of the origin with $S_x = 2$ and $S_y = 2$. A vertex of the object is $A(4, 4)$ using the composite matrix method find the new coordinates of A .

$$\Rightarrow \text{Given: } F = (2, 2)$$

$$A = (4, 4)$$

$$S_x = S_y = 2$$

Step 1: translate fixed point F to origin.

$$T_1 = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 3: } T_2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = T_2 S T_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -4 \\ 0 & 2 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = T_2 S T_1 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A' = (C, C) \text{ and}$$

Q.6 Find the image of point $P(3, 4)$ after reflection about the X -axis using matrix multiplication.

$$P = [3, 4]$$

$$R_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow P' = R_x P = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Q.7 Find the image of point $P(3, 4)$ after reflection about the Y -axis using matrix multiplication.

$$P = [3, 4]$$

$$R_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$P' = R_y P = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$P' = [-3, 4] \text{ and}$$

Q.8 Reflect the point $P(2,5)$ about the line $Y = X$ using the homogeneous reflection matrix.

$$\Rightarrow R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

$$P' = R \times P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \boxed{P'(5,2)}$$

Q.9 A 3D point $P(2,3,4)$ is reflected about the xy -plane ($Z=0$). find the new coordinates using matrix multiplication.

$$R_{xy} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

$$P' = R_{xy} P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -4 \\ 1 \end{bmatrix} = \boxed{P'(2,3,-4)}$$

Q.10 Reflect point $P(1,2)$ about the origin in 2D using matrix multiplication, and verify the result by reflecting about the x -axis followed by the y -axis.

